XEZIM

--module-timescale User Guide

Assign module timescales from the command line without overriding source-explicit settings.

Rule: A module-local timeunit/timeprecision or a `timescale placed directly before that module is protected. A module that only inherits an earlier `timescale may be changed by --module-timescale.

Syntax

```
# All eligible modules
xezim --module-timescale 1ns/1ps design.sv

# Selected modules
xezim --module-timescale module_c,module_d=10ns/1ns design.sv

# Multiple groups
xezim \
  --module-timescale 1ns/1ps \
  --module-timescale module_c=10ns/1ns \
  --module-timescale module_d,module_e=1ps/1ps \
  design.sv
```

Example

```
`timescale 1ns/1ps
module module_a;
  initial #1 $display("A");
endmodule

`timescale 10ns/1ns
module module_b;
  initial #1 $display("B");
endmodule

module module_c;  // inherits B in normal LRM behavior
  initial #1 $display("C");
endmodule

module module_d;
  timeunit 100ps / 1ps;
  initial #1 $display("D");
endmodule
xezim --module-timescale module_c=1ps/1ps design.sv
```

Module	Effective timescale	Why
module_a	1ns/1ps	Protected direct `timescale
module_b	10ns/1ns	Protected direct `timescale
module_c	1ps/1ps	Named command-line assignment
module_d	100ps/1ps	Protected local timeunit

Precedence

local timeunit/timeprecision > direct `timescale > named assignment > global assignment > inherited `timescale > simulator default

Useful checks

```
xezim --report-timescales design.sv
```

- Repeat --module-timescale for different module groups.
- Conflicting values for the same module are errors.
- The option applies to module definitions, so all instances use the same assigned value.
- Use a fresh `timescale before a module when its source setting must be protected.